

Selective HER Activation of Monolayer CrCl_3 by Transition-Metal Functionalization

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Supporting Information Available

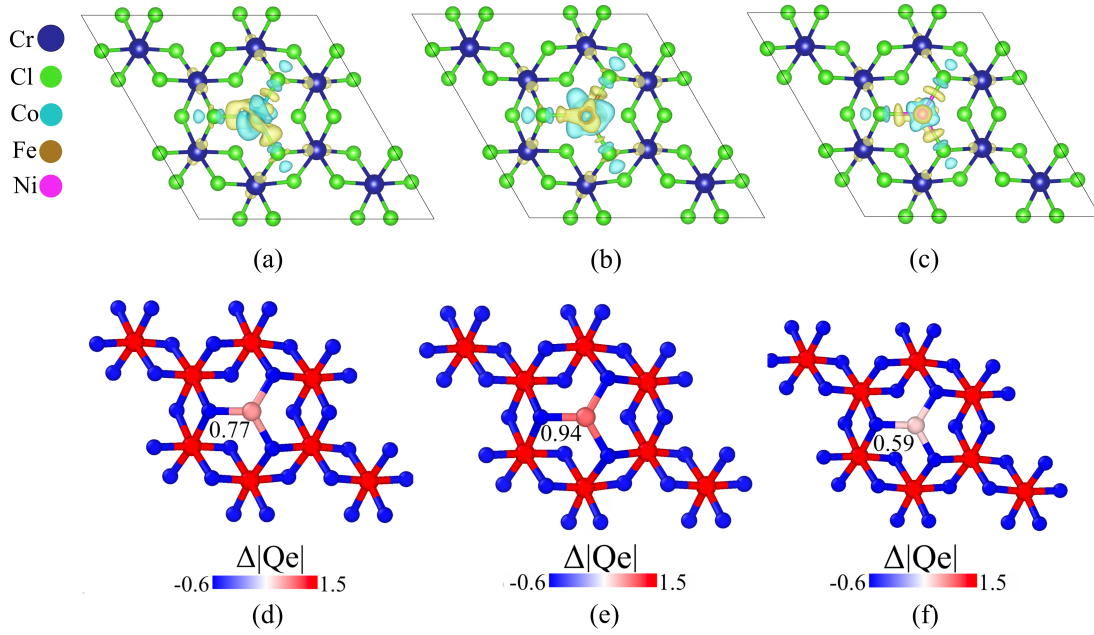


Fig. S1. Top views of the charge-density-difference isosurfaces for (a) Co-, (b) Fe-, and (c) Ni-adsorbed CrCl₃, respectively, plotted at an isosurface value of $\delta = 5^{-3}$ eV/Å³. The two colors denote regions of charge accumulation and charge depletion induced by adatom adsorption. (d-f) Top views of the corresponding Bader charge analysis for Co-, Fe-, and Ni-adsorbed CrCl₃, respectively. The values indicated in each panel represent the magnitude of the charge transfer associated with the adsorbed transition-metal atom, in units of $|e|$.

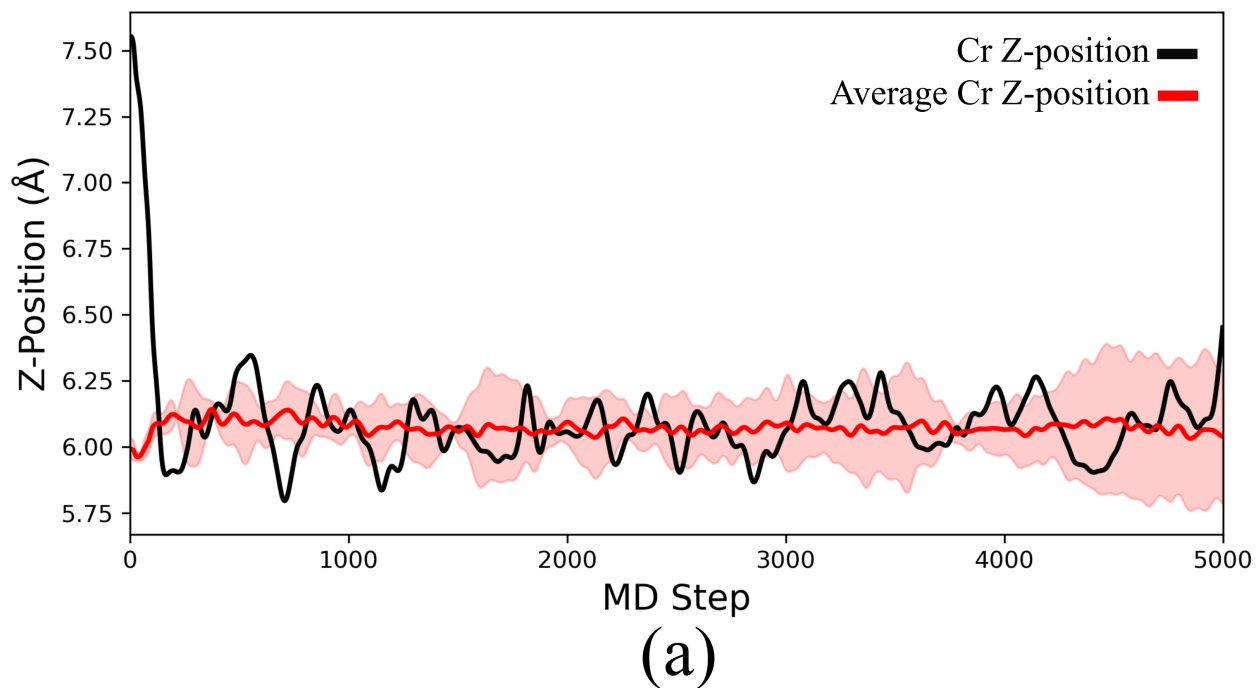


Fig. S2. Time evolution of the total energy (blue curves, left axis) and instantaneous temperature (red curves, right axis) obtained from the AIMD simulations performed at 300 K for the CrCl_3 monolayer with Cr adsorbed at the hollow site.

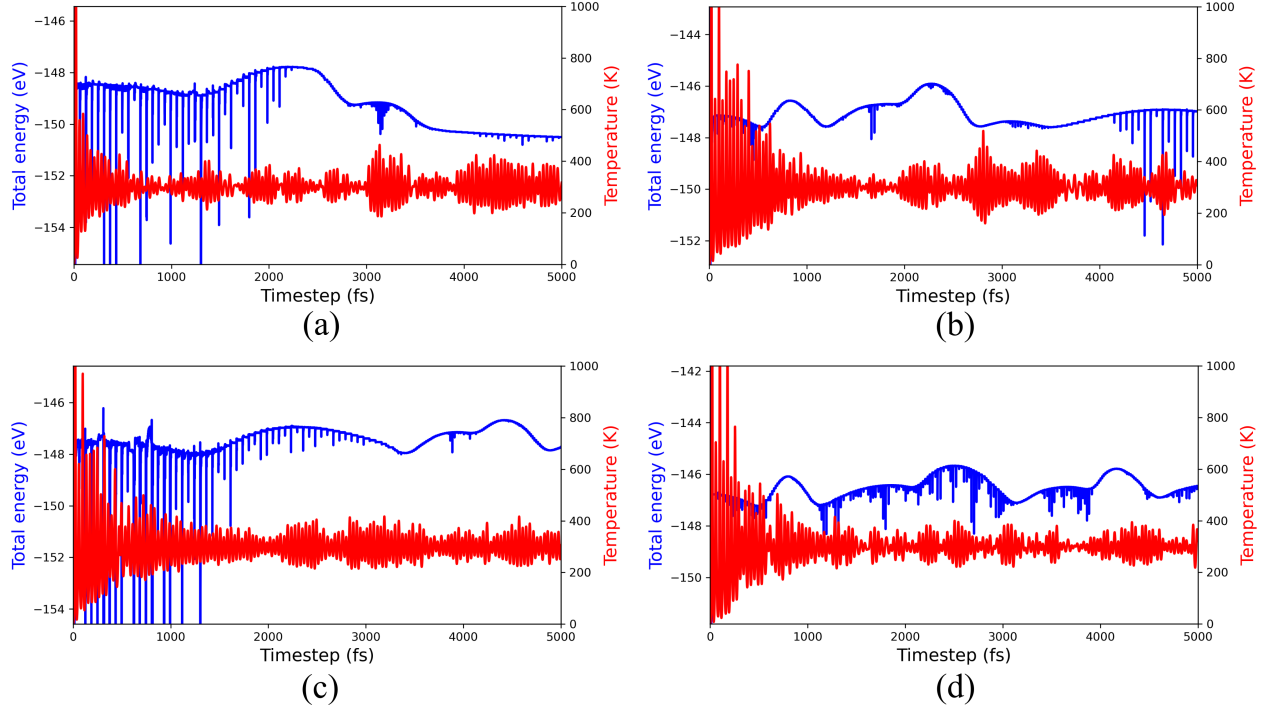


Fig. S3. Time evolution of the total energy (blue curves, left axis) and instantaneous temperature (red curves, right axis) obtained from the AIMD simulations performed at 300 K for the CrCl_3 monolayer with adsorbed (a) Cr, (b) Co, (c) Fe, and (d) Ni atoms at the hollow site. The results confirm the thermal equilibration of all systems during the 5 ps simulation window.

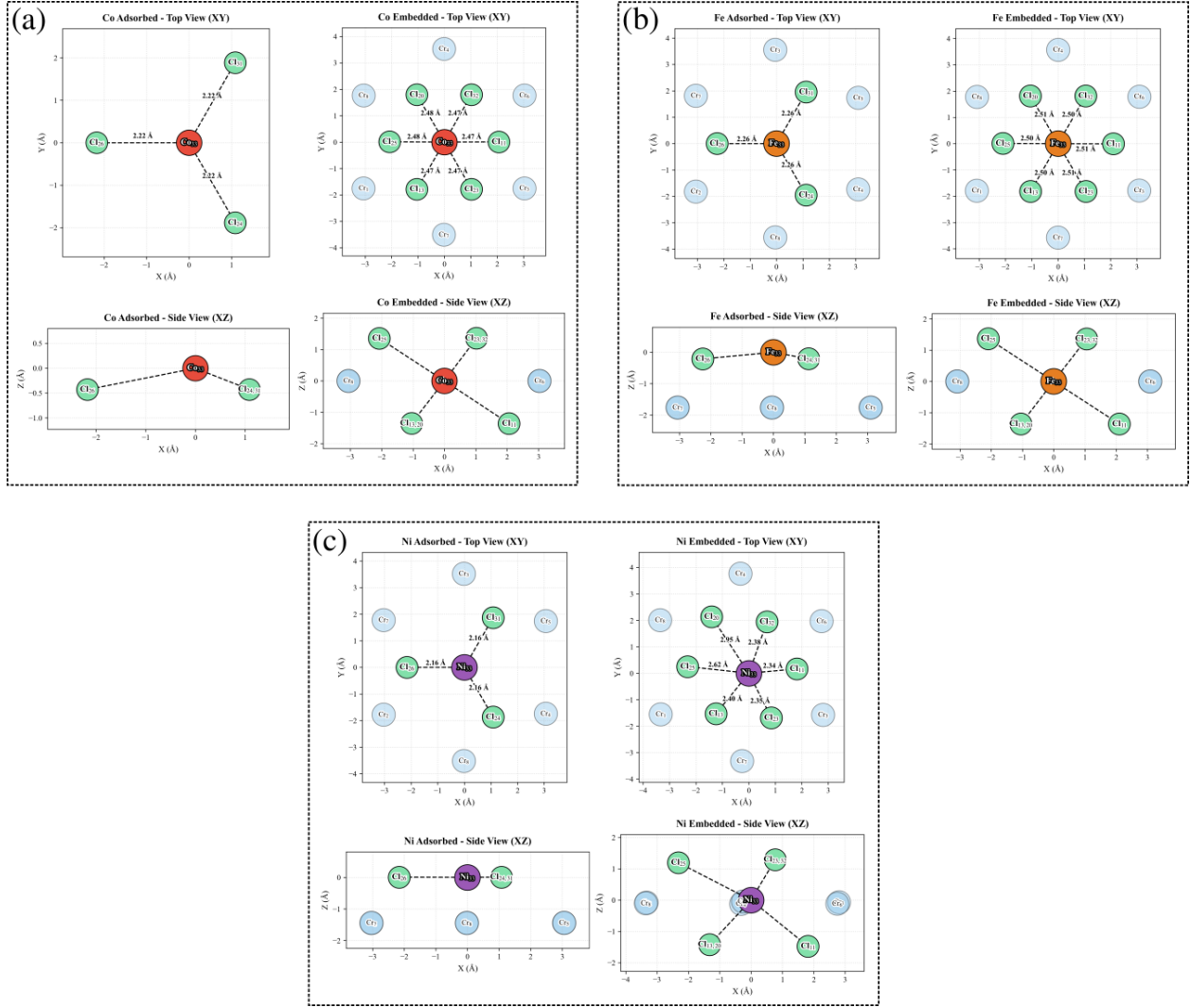


Fig. S4. Schematic top (XY) and side (XZ) views of the optimized local coordination geometries and TM-Cl bond lengths (Å) for (a) Co-, (b) Fe-, and (c) Ni-functionalized monolayer CrCl₃ in both adsorbed (hollow site, 3-fold coordinated) and embedded (6-fold octahedrally coordinated) configurations. Light blue, light green, and red/orange/purple spheres represent Cr, Cl, and transition-metal (Co, Fe, Ni) atoms, respectively.

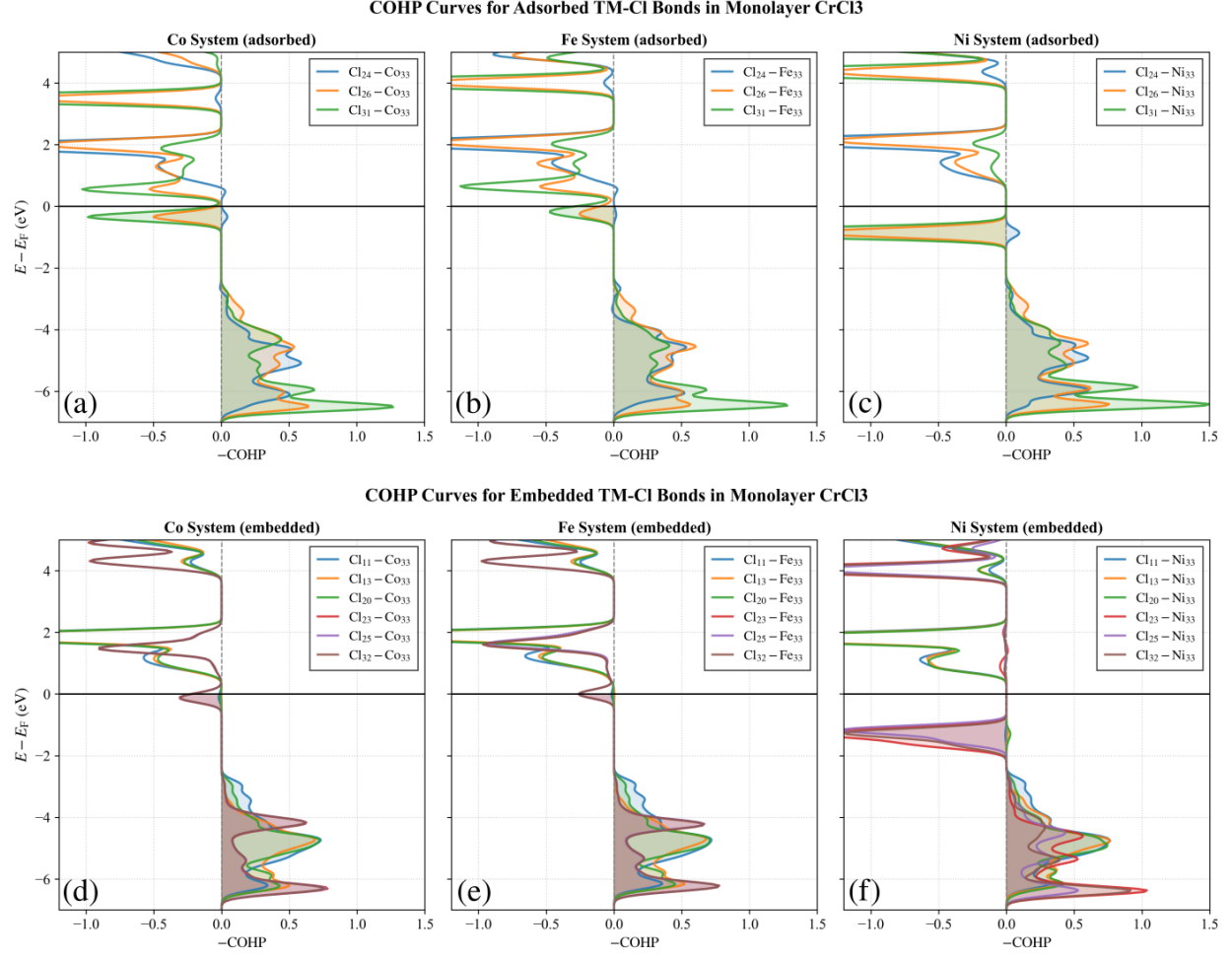


Fig. S5. Crystal Orbital Hamilton Population ($-\text{COHP}$) energy curves calculated for the individual TM-Cl (TM = Co, Fe, Ni) chemical bonds in monolayer CrCl₃. (a–c) $-\text{COHP}$ diagrams for the 3-fold coordinated adsorbed Co, Fe, and Ni systems, respectively. (d–f) $-\text{COHP}$ diagrams for the 6-fold octahedrally coordinated embedded Co, Fe, and Ni systems, respectively. The horizontal black line indicates the Fermi level ($E - E_F = 0$ eV). Shaded regions below E_F highlight bonding orbital contributions ($-\text{COHP} > 0$), while curves extending into negative values denote anti-bonding states.

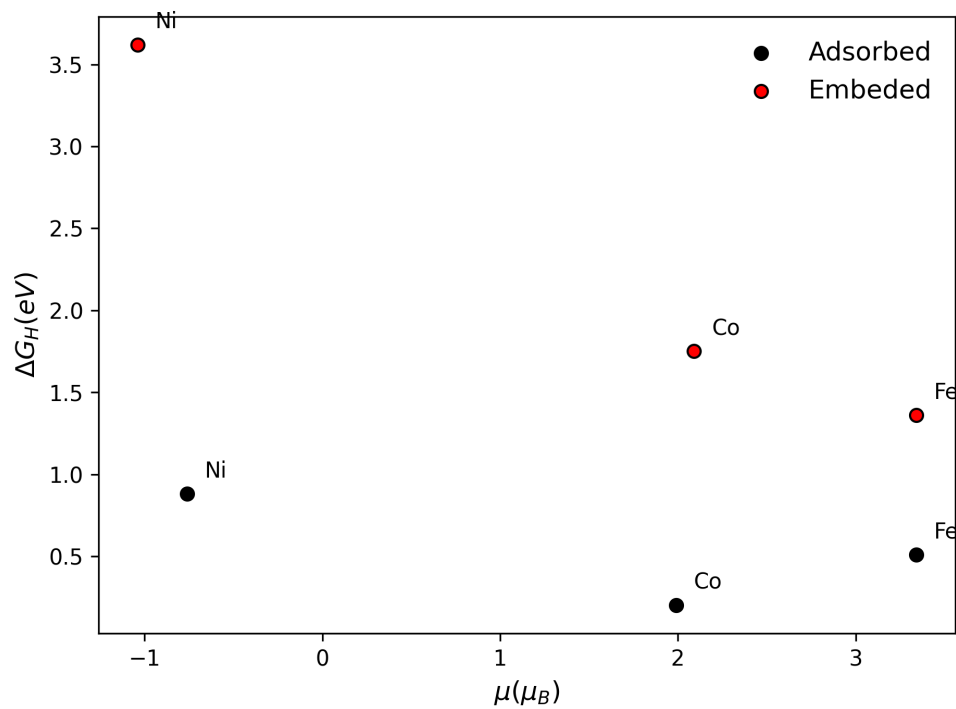


Fig. S6. Relationship between the magnetic moment (μ_B) of the transition-metal atoms and the corresponding hydrogen adsorption free energy (ΔG_H) for Co-, Fe-, and Ni-functionalized CrCl_3 monolayers in adsorbed and embedded configurations.